REDNet: Reliable Evidential Discounting Network for Multi-Modality Medical Image Segmentation

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***Abstract*— In the field of computer-aided diagnosis, particularly for tumor diseases, segmentation is a prerequisite and primary step. Multi-modality images become essential for achieving accurate segmentation, which offer critical insights beyond the limitations of single-modality data. However, different modalities and images may suffer from different types of data imperfection, such as intensity non-uniformity, motion artifact, and low quality due to hardware limitations, which challenge image segmentation algorithms. To address this challenge, we propose a Reliable Evidential Discounting Network (REDNet), which is composed of three main modules: 1) the Intra-modality Consistency Evaluation Module (ICEM) measuring the data cohesion within the same modality; 2) the Cross-modality Difference Aggregation Module (CDAM) identifing data discrepancy across modalities; 3) the Discounting Fusion Module (DFM) processing the multi-modality evidence by applying discounting strategies to fuse the data. This approach maintains segmentation accuracy by effectively integrating multi-modality evidence, while discounting the influence of lower-quality data, ensuring reliable results despite the presence of image imperfections. We evaluated REDNet on two distinct datasets, BRATS2021 and an in-house pancreas dataset from Changhai Hospital. REDNet outperforms other methods, particularly in scenarios with imperfect image sources, and achieves reliable results in multi-modality tumor segmentation.**

***Index Terms*— Reliable segmentation, multi-modality data, evidential learning, uncertainty.**

I. Introduction

MEDICAL image segmentation aims to automatically and accurately delineate organs and tumors, serving as an essential component for computer-aided diagnosis [1], [2], [3], [4]. Single-modality medical images often lack the detail needed for reliable diagnoses, especially for tumors which prompting researchers to rely on multi-modality imaging to achieve enhanced segmentation accuracy. Researchers have designed different networks to improve segmentation performance on multi-modality [5], [6], [7].

The performance of existing multi-modal medical image segmentation frameworks is heavily dependent on high-quality data from various modalities. In clinic scenarios, images often suffer from various types of imperfection due to uneven illumination, patient movement, hardware limitations, and other unexpected factors [8], [9], [10]. Models often perform poorly when faced with imperfection in one or more modalities. Furthermore, many researchers have proposed uncertainty approaches to reduce the impact of imperfection on medical image segmentation models. Methods such as shadowed sets [11], ensemble [12] and subjective logic [13] have shown potently of tackling imperfection. However, Karimi and Gholipour [14] shows that uncertainty methods are less effective when there is a substantial discrepancy between test and training data, which challenges the generalizability. Many existing models may fail to detect imperfection in the data and cannot dynamically adjust based on data quality to reduce the effects of flawed data.

In this work, we propose a reliable evidential discounting network, named REDNet, which not only dynamically adjusts the contribution of modalities based on their real-time quality for better segmentation but also actively reduces the impact of imperfection in multi-modality. REDNet is composed of three main modules. The Intra-modality Consistency Evaluation Module (ICEM) refines features within modalities by contrasting test against training data, while the Cross-modality Difference Aggregation Module (CDAM) extracts cross-modality insights by observing data discrepancy across modalities. Unlike previous methods which apply static weights, the Discounting Fusion Module (DFM) offers an innovative approach that varies the level of discounting in response to the observed imperfection in each modality and fuse the data from different modalities, thereby ensuring reliable segmentation under diverse conditions.

We conduct experiments to evaluate the effectiveness of the proposed REDNet on the BRATS2021 challenge dataset [15] and in-house pancreas dataset. Our model performs well

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against both conventional methods and uncertainty-based methods. Our model demonstrates strong performance in introducing the impact of imperfection to the data as well as handling the inherent imperfections present in medical imaging data. The main contributions of this study are as follows:

- We propose a reliable evidential discounting network (REDNet). This framework is designed to dynamically calibrate the influence of each modality by assessing their real-time quality and reducing the impact of imperfection.
- We present three modules: ICEM identifying consistency between test and training data, CDAM evaluating data discrepancy across modalities and DFM fusing multi-modal data by employing discounting methodologies. DFM also customizes the impact of each modality based on data quality at a pixel level.
- We validate the proposed method on two distinct datasets, BRATS2021 and an in-house pancreas dataset, comparing it with both classic and uncertainty-based methods. Our proposed framework performs well in both normal scenarios and imperfection scenarios.

II. Related Works

A. Multi-Modal Medical Image Segmentation

Many researchers have proposed multi-modal fusion methods to fully utilize the different information contained in multi-modal data. The current multi-modal fusion methods can be mainly divided into early fusion [7], [16], [17], intermediate fusion [18], [19], and decision fusion [20], [21]. Early fusion methods mainly involve combining the raw data at the level of the original data. The typical method is to simply concatenate or add the data and then input it into a neural network. Chang et al. [17] propose DPAFNet, based on dual-path (DP) module and multi-scale attention fusion (MAF) module. By concatenating multimodal inputs, the network ensures precise tumor segmentation. Intermediate fusion is widely used in multi-modal fusion methods, because it allows for the combination of information from different modalities in a more meaningful and interpretable way. For example, Xu et al. [18] innovate a knowledge-guided multimodal fusion methodology, which uses multi-phase self-attention module to make full use of cross-modal information. Zhang et al. [19] introduce a spatial aggregation module to encourages per-pixel interactions between different phases. Decision fusion achieves the final outcomes by integrating results from multiple phases through methods, such as averaging. Guo et al. [20] employs averaging to merge segmentation outputs from multimodal CNNs to improve segmentation accuracy.

Although these methods offer high precision, they often rely on empirical design to extract multi-phase information. Yet, their results are not always explicitly interpreted, depriving clinicians of effective insights into the models reliability in uncertain scenarios.

B. Uncertainty-Based Medical Image Segmentation

Using uncertainty in segmentation helps identify regions where the model's predictions have a lower degree of confidence. Uncertainty-based methods, as mentioned in [22], can be divided into probabilistic-based, ensemble-based, and evidence-based methods. Probabilistic-based [23], [24] methods capture ambiguity in the data by learning a distribution over segmentations, rather than producing a single deterministic output. Through this technique, the model can provide both the most likely segmentation result along with a measure of uncertainty, indicating the level of confidence for each pixel. Ensemble-based methods [12], [25] aim to quantify predictive uncertainty in deep learning models. By training multiple neural networks independently from different initializations, this method constructs an ensemble of models. Evidence-based methods [7], [26] use an evidential layer which can quantify the evidence-based uncertainty using subject logic (SL) [13] and Dempster-Shafer theory (DST) [27], [28]. Zou et al. [7] propose a segmentation network which can generate robust segmentation results and reliable uncertainty using subject logic. Huang et al. [29] introduce an Evidential Segmentation (ES) module that not only achieves improved segmentation accuracy in 3D PET/CT imaging but also generates uncertainty maps that effectively quantify classification uncertainty.

Although these methods have made progress in quantify uncertainty, they fall short in resisting noise. They can effectively capture areas of uncertainty when data corruption is minimal, but fail to accurately measure uncertainty and yield good segmentation results when the data is completely compromised.

III. Methodology

The overall framework of REDNet is shown in Fig.1. It is composed of evidential-based uncertainty estimation module, Intra-modality Consistency Evaluation Module (ICEM), Cross-modality Difference Aggregation Module (CDAM), and Discounting Fusion Module (DFM). We are initially provided with a set of multi-modality data, denoted as $\mathcal{X} = \{x^i\}_{i=1}^{M}$, where M represents the total number of modalities, each $x^i \in \mathbb{R}^{D,H,W}$ represents an original input from a specific modality, $\mathcal{X}$ is the set of all modalities involved and D, H, W is the size of the image. The proposed framework processes 3D medical images, and all modules operate on volumetric data to preserve 3D contextual information. For each individual modality, an evidence representation e^i is then computed through a Subjective Logic Network:

$$e^i = F^i(x^i), \tag{1}$$

where $e^i \in \mathbb{R}^{C,D,H,W}$ and C is the class number. Each modality possesses its unique extractor F^i. Then, the obtained evidence from each modality is conveyed into the ICEM and CDAM to extract single modality information and cross-modality information, denoted as d_{in} and d_c, respectively.

Next, the data is introduced to the Discounting Fusion Module. Using the information received from ICEM and CDAM, the credibility of each pixel is evaluated within each modality, leading to a discount on the evidence e^i acquired in Eq. (1). The final process constitutes using the Dempster-Shafer (DS) Evidence Theory for an integrated fusion, resulting in a final segmentation. For convenience, the symbols used in this paper are summarized in Table I.

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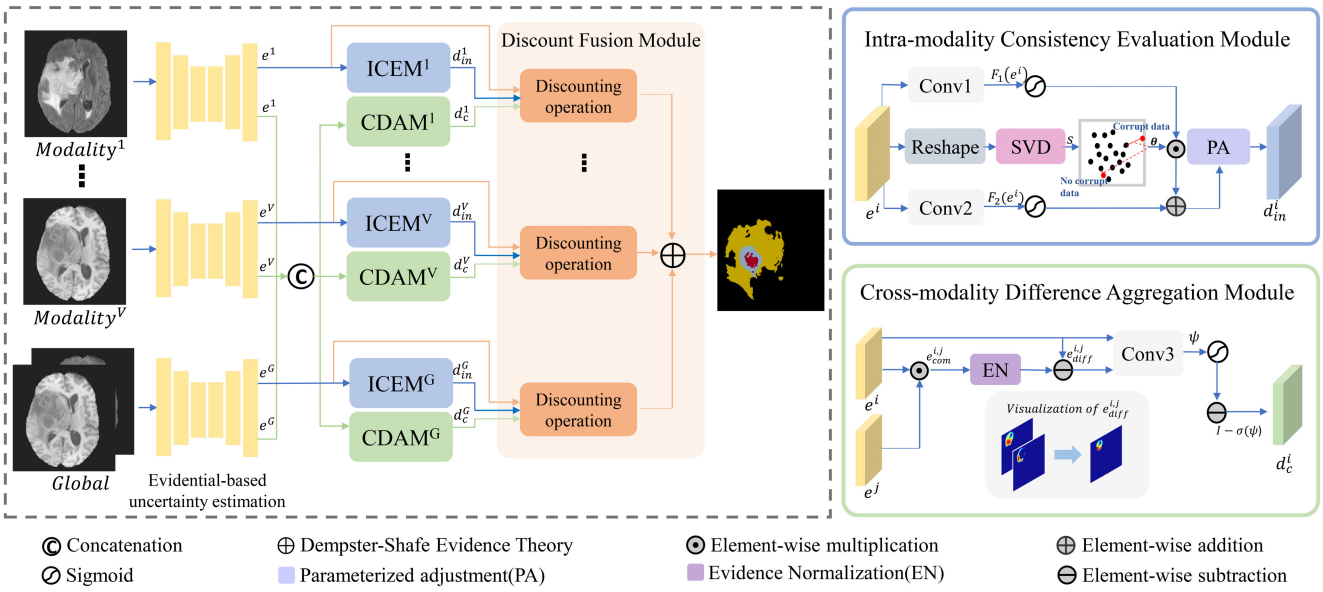


Fig. 1. This framework performs reliable multi-modality image segmentation. Treat all modality data as input data, where 'Global' denotes the concatenation of all modalities. Different imaging modalities are processed individually through evidential-based uncertainty estimation to extract features and generate evidence e^i. These are then refined by Intra-modality Consistency Evaluation Module (ICEM) and Cross-modality Difference Aggregation Module (CDAM) for quality detection. The Discounting Fusion Module (DFM) processes the multi-modality evidence by applying discounting strategies to fuse the data.

TABLE I
DEFINITION OF SYMBOLS USED IN THIS PAPER

Symbol	Definition	Symbol	Definition
$\mathcal{X}$	Set of all input modalities	S^i_{train}	Stored singular values of training data in epoch i
x^i	Input data from i-th modality ($x^i \in \mathbb{R}^{D,H,W}$)	θ_t	Consistency measure between test data and training data
D, H, W	Depth, height, and width of 3D medical images	α	Learnable parameter in ICEM
e^i	Evidence representation for modality i	$e^{i,j}_{\text{com}}$	Commonality evidence between modalities i and j
C	Number of classes	γ	Normalization factor
F^i	Layer of network for modality i	e_{diff}	Differential evidence between modalities i and j
$b^n_{i,j,k}$	Belief mass for class n at voxel (i,j,k)	ψ	Cross-modality descriptor
$u_{i,j,k}$	Uncertainty at voxel (i,j,k)	b^d_k	Discounted belief for class k
$a^n_{i,j,k}$	Dirichlet parameter	u^d	Discounted uncertainty
S	Dirichlet strength	$m(X)/m_d(X)$	Opinion/Discounted opinion
U, Σ, V	SVD matrices	$m^{\text{merge}}_d(X)$	Aggregated belief after discounting and fusion
s	Singular values vector ($s = \text{diag}(\Sigma)$)	$\lambda_p, \lambda_d, \lambda_m$	Loss weighting parameters

A. Evidential-Based Uncertainty Estimation

Based on Dempster-Shafer Evidence Theory (DST) [30], a framework for Subjective Logic (SL) [13] is developed that specifically correlates belief and uncertainty with the parameterization of the Dirichlet distribution. Subjective Logic (SL) associates the parameters of the Dirichlet distribution with the belief distribution, where the Dirichlet distribution can be considered as the conjugate prior of the categorical distribution [31]. SL assigns a belief mass b and an uncertainty mass to each phase, presenting the segmentation and ensuring trustworthy results. Each image pixel for the s-th phase prediction is assigned a belief mass and uncertainty, with the following constraint:

$$\sum_{n=1}^{C} b^n_{i,j,k} + u_{i,j,k} = 1, \tag{2}$$

where C denotes the class number, $b^n_{i,j,k} \geq 0$ and $u_{i,j,k} \geq 0$ are the belief mass belonging to the n-th class and the uncertainty. The coordinates $(i, j, k) \in (D, H, W)$ denote the position of $b^n_{i,j,k}$ and $u_{i,j,k}$ for the prediction. Assuming constant base rate, the evidence $e^n_{i,j,k}$ are associated with the Dirichlet parameters $a^n_{i,j,k}$ by:

$$a^n_{i,j,k} = e^n_{i,j,k} + 1, \tag{3}$$

where $e^n_{i,j,k} \geq 0$ is obtained directly from the backbone network with a softplus activation function. Thus, the belief mass and the uncertainty can be formulated as:

$$b^n_{i,j,k} = \frac{e^n_{i,j,k}}{S} = \frac{a^n_{i,j,k} - 1}{S} \quad \text{and} \quad u_{i,j,k} = \frac{C}{S}, \tag{4}$$

where $S = \sum_{n=1}^{C} a^n_{i,j,k} = \sum_{n=1}^{C} (e^n_{i,j,k} + 1)$ denotes the Dirichlet strength. A greater amount of evidence $e^n_{i,j,k}$ leads

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to a higher probability being assigned to the corresponding outcome. Conversely, when there is a lack of evidence, the resultant belief is less certain, thereby increasing the measure of uncertainty.

B. Intra-Modality Consistency Evaluation Module

As seen in Fig. 1, this module focuses on intra-modality analysis and recalibrates each pixel's information. By quantifying the data cohesion between the test data and training set distributions, it refines pixel-level characteristics and adapts feature representation to ensure robust against imperfection. For a more refined characterization of the feature space, we compute the spectrum of the feature, which can be computed using singular value decomposition (SVD). As described in Sec. III-A, for each modality, we obtain the evidence matrix denoted as $e \in \mathbb{R}^{C,D,H,W}$ Then it is reshaped into $e \in \mathbb{R}^{D*H*W,C}$, followed by the computation of its SVD:

$$e = U \Sigma V^T, \tag{5}$$

where U and V are orthonormal matrices and the diagonal matrix Σ is comprised of the singular values of e, with these diagonal elements being non-negative. The vector of singular values, $s = \text{diag}(\Sigma)$, reflects the principal components of the feature.

Inspired by [14], we design a way to quantify the consistency between the singular values s with training-data. We compute the singular values of all training data and store them as S^i_{train} in a training epoch i. Given an evidence e in test or in the next epoch, we calculate its singular values s. The consistency between the given data and training-data can be define as:

$$\theta = \min\left(\frac{\|s - s_t\|_1}{C}\right), \quad \forall s_t \in S^i_{\text{train}}, \tag{6}$$

where C is the class number, s_t is defined as the singular values of training data, s represents the singular values of test-time input data or data from subsequent epochs. θ reflect diversity: a low value indicates high consistency, while a high value indicates less consistency.

To precisely capture per-pixel modality-specific information, we have added convolutional layers to complement the consistency metric, thus representing modality details more effectively:

$$f' = \sigma(F_1(e)) * \theta_t + \sigma(F_2(e)), \tag{7}$$

where $F_i(\cdot)$ is convolution layers and $\sigma(\cdot)$ represents the sigmoid activation function.

Then, the output of ICEM block can be obtained by parameterized adjustment:

$$d_{in} = \sigma(\alpha - f'), \tag{8}$$

where α is a learnable parameter, d_{in} can adjust the impact of modalities based on the imperfection situation: the high d_{in} discount the modality's contribution to fusion, Conversely, low divergence retains the modality's influence, as d_{in} approaches one. This ensures feature fusion is finely tuned to the quality of the data.

In summary, this module refines modality-specific information at the pixel level and dynamically enhances features by incorporating the diversity θ, permitting the module to adapt based on the modality data's quality.

C. Cross-Modality Difference Aggregator Module

To fully exploit the information between multimodal data, we further propose a cross-modality diversity aggregator module. When significant discrepancies are detected between a specific modality and others, it suggests potential unreliability in that modality (e.g., due to noise, artifacts, or modality-specific biases). To mitigate this, the module adaptively reduces the weight assigned to the unreliable modality in the final prediction. This ensures that regions with inconsistent or noisy signals contribute less to the segmentation outcome, thereby enhancing overall robustness. This module integrates information from different modalities to enhance system robustness and precision. It reduces the impact of areas with significant discrepancies where different modalities provide different information. This allows the model to focus more on different information across modalities and minimizes imperfection interference.

Let $\mathcal{X} = \{x^i\}_{i=1}^M$ denote the set of input modalities, where x^i is the $i-th$ modality image. For two distinct modalities $x^i, x^j \in \mathcal{X}$ and $j \neq i$, let e^i and e^j be their corresponding evidence. We define the differential evidence $e^{i,j}_{\text{diff}}$ as the difference in evidence between modalities i and j, which can be articulated through the following expressions:

$$\begin{aligned} e^{i,j}_{\text{com}} &= e^i \odot e^j, \\ \gamma &= max(e^j, c), \\ e^{i,j}_{\text{diff}} &= e^i - e^{i,j}_{\text{com}}/(\gamma + \epsilon), \end{aligned} \tag{9}$$

where $e^i, e^j \in (C, D, H, W)$, $\odot$ is element-wise multiplication, $e^{i,j}_{\text{com}}$ denotes the commonality between e^i, e^j at matching spatial locations. $max(e^j, c)$ represents taking the maximum value across the class channels. Parameter γ serves as a normalization factor. The differential evidence, e_{diff}, is calculated by subtracting the normalized common evidence from e^i, and γ ensures that the minimum value of the resulting difference is zero. The ϵ term is infinitesimal value to avoid singularity (such as $\epsilon = 1e-10$). This method ensures that e_{diff} highlights the distinct features of e^i relative to e^j.

Then, we combine e^i with the differential evidence $e^{i,j}_{\text{diff}}$ from all other modalities:

$$\begin{aligned} f^i_{\text{concat}} &= \text{Concat}\left(e^i, \left\{e^{i,j}_{\text{diff}} \mid j \neq i\right\}\right), \\ \psi &= F(f^i_{\text{concat}}). \end{aligned} \tag{10}$$

This operation obtains a fused feature, f^i_{concat}, which captures the unique features of e^i and highlights its differences from other modalities. $F(\cdot)$, the convolution layers, then processes this combined set of features to create a Cross-Modality Descriptor (ψ), enhancing the network's ability to detect how much a modality differs from others. We calculate cross modality weights d_c in modality i, using:

$$d_c = I - \sigma(\psi), \tag{11}$$

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where I is an all-ones matrix representing that the maximum weight is one. The lesser the difference, as indicated by a smaller $\sigma(\psi)$, the greater the weight attributed. Eq. 11 translates the difference detection into weights for downstream tasks. This step is critical for ensuring that the fusion process is both interpretable and adaptive to input-specific modality relationships. A large discrepancy between modalities i and j increases ψ, reducing d_c. To apply more discounts in the fusion process. Small discrepancies make the d_c closer to 1, resulting in fewer discounts. In this way, it gives priority to those features that are more reliable in multimodal analysis, while reducing the impact of imperfection.

D. Discounting Fusion Module

In our framework, evidence is dynamically adjusted based on the quality of the modality, to express degrees of trust in an information source. This adjustment occurs before the fusion stage. Let variable X be a belief, and let opinion $m(X) = (b, u)$. The framework mathematically modulates the opinion on X by applying a trust-discounting operator $m_d(X)$:

$$\begin{aligned} b_k^d &= d_k * b_k, \\ u^d &= 1 - \sum_C d_k * b_k, \end{aligned} \tag{12}$$

where b_k^d and u^d represent the discounted belief and uncertainty, respectively. d^k represents the discount factor for the kth category. This discounting serves to diminish the influence of less reliable information sources.

Trust discounting is a technique that scales beliefs according to the assessed reliability of the information source. By weighting the information in this way, we can achieve a more detailed evaluation that considers both the content of the evidence and the credibility of the information source.

Building upon discounting operator, we integrate the d_{in} and d_c matrices, indicative of intra-modality and cross-modality information, respectively, to calculate the discount factor. This integration allows for a more granular calibration of the discounting:

$$d_k = \frac{(\alpha * d_{in}) + (\beta * d_c)}{(\alpha + \beta)}, \tag{13}$$

where α and β are hyperparameters, both of them are set to the value of 1. This adjustment, taking into account both intra-modality and cross-modality information, enables the dynamic tuning of evidence based on the quality of the modality before advancing to the fusion stage. By refining the degree of uncertainty, it allows useful information to play a more significant role while reducing the impact of imperfection data. The discounting strategy is adaptive to data quality: for high-quality inputs with minimal imperfections, the computed discount factor d approaches 1, effectively preserving the original segmentation. This ensures that the framework defaults to standard predict when data reliability is assured, avoiding unnecessary modifications.

Having applied the discounting method to evaluate the credibility of the evidence, we proceed to the data fusion phase. Using Reduced Dempster's Combination Rule [31], two mass functions m_1 and m_2 from independent modalities are merged:

$$b_k = \frac{1}{1-C}\left(b_k^1 b_k^2 + b_k^1 u^2 + b_k^2 u^1\right), u = \frac{1}{1-C} u^1 u^2, \tag{14}$$

where $C = \sum_{i \neq j} b_i^1 b_j^2$ represents the conflict level between the two mass sets and $\frac{1}{1-C}$ acts as the normalizing factor.

Then, the joint opinion m, derived by merging opinions from different modalities, can be expressed as follows:

$$m = m^1 \oplus m^2 \oplus \cdots \oplus m^{|\mathcal{X}|}. \tag{15}$$

After discounting the evidence using the trust-discounting operator, we proceed with the fusion of the resulting discounted evidence:

$$m_d^{merge}(X) = \oplus_{\forall x \in \mathcal{X}}(m_d^x(X)), \tag{16}$$

where $m_d^{merge}(X)$ represents the aggregated belief function and is the final result.

E. Loss Function

The loss function in the training procedure consists of phase-wise and fusion-wise losses, which can be defined as:

$$\mathcal{L} = \overbrace{\frac{\lambda_p}{|\mathcal{X}|}\sum_{x \in \mathcal{X}} \mathcal{L}_t + \frac{\lambda_d}{|\mathcal{X}|}\sum_{x \in \mathcal{X}} \mathcal{L}_d}^{\text{phase-wise}} + \underbrace{\lambda_m \mathcal{L}_m}_{\text{merge-wise}}, \tag{17}$$

where $\lambda_p, \lambda_d, \lambda_m$ are uniformly set to 1. $\mathcal{L}_t$ is proposed in [31] to quantify the voxel-wise uncertainty and belief mass. It can be defined as:

$$\mathcal{L}_t = \mathcal{L}_{\text{ice}} + \lambda_p \mathcal{L}_{\text{KL}} + \lambda_s \mathcal{L}_{\text{dice}}. \tag{18}$$

$\mathcal{L}_{\text{ice}}$ is designed for the imbalance of brain tumor and $\mathcal{L}_{\text{KL}}$ guarantees that incorrect labels get less evidence even shrinking to 0. Both $\mathcal{L}_d$ and $\mathcal{L}_m$ represent the Dice loss, which respectively impose constraints on the outcomes following discounting and on the results fusion, ensuring consistency and accuracy in the segmentation process.

IV. Experiment

A. Experimental Setup

1) Dataset: We evaluated REDNet on both external dataset and in-house pancreas dataset. The external dataset consists of 1251 MRI cases can be openly obtained from the Multimodal Brain Tumor Segmentation Challenge 2021 (BraTS2021) [15]. The BraTS image segmentation challenge started in 2012 and has been conducted yearly as a crucial component of the MICCAI [32]. The BraTS dataset is a widely used benchmark for brain tumor segmentation research. All the imaging datasets have been segmented manually, by one to four raters, following the same annotation protocol, and their annotations were approved by experienced neuro-radiologists. For each scan, there are four modalities: T1-weighted (T1), post-contrast T1-weighted (T1Gd), T2-weighted(T2), and T2 Fluid Attenuated Inversion Recovery (FLAIR). Annotations of the 1251 scans include the GD-enhancing tumor (ET-label 4), the peritumoral edema (ED-label2), and the

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TABLE II
THE PERFORMANCE OF DIFFERENT MODELS IN BRATS21 DATASET AND IN-HOUSE DATASET(† MEANS UNCERTAINTY-BASED MODELS). THE TOP-2 RESULTS ARE IN RED AND BLUE

models	A. Brats (Brain MRI)								B. Inhouse (Pancreas CT)					
	DICE				HD95				DICE			HD95		
	ET	TC	WT	Mean	ET	TC	WT	Mean	Pancreas	Tumor	Mean	Pancreas	Tumor	Mean
AttentionU [35]	0.8075	0.8340	0.9015	0.8477	16.49	10.53	11.53	12.85	0.8464	0.7561	0.8012	13.05	16.58	14.81
UNetr [36]	0.8410	0.8653	0.9133	0.8732	13.80	9.26	10.64	11.23	0.6847	0.5581	0.6214	37.13	31.59	34.36
SegResNet [37]	0.8407	0.8931	0.9237	0.8858	14.79	6.80	6.30	9.30	0.8590	0.7008	0.7799	15.67	23.14	19.41
UNet++ [38]	0.8450	0.8880	0.9210	0.8847	15.18	7.29	9.34	10.60	0.8961	0.8115	0.8538	8.49	12.19	10.34
CKD-TransBTS [33]	0.8850	0.9016	0.9333	0.9066	5.93	6.54	6.20	6.22	N/A	N/A	N/A	N/A	N/A	N/A
†UE [12]	0.8151	0.8586	0.9003	0.8580	13.23	9.38	9.05	10.55	0.8296	0.6505	0.7401	20.28	36.55	28.42
† DU [39]	0.8397	0.8845	0.9191	0.8811	11.44	6.77	6.77	8.33	0.8378	0.6933	0.7656	18.53	28.64	23.59
† TBraTs [7]	0.8466	0.8846	0.9283	0.8865	13.36	7.18	6.41	8.98	0.8853	0.8192	0.8523	11.44	8.76	10.1
ours	0.8532	0.8963	0.9343	0.8946	12.39	6.07	5.71	8.05	0.9035	0.8411	0.8723	6.94	8.13	7.54

In B.Inhouse, there is no comparative data because the CKD-TransBTS [33] method employs prior medical knowledge of the brain.

necrotic and non-enhancing tumor core (NCR/NET -label1). Following [33], we divided the 1251 scans into 834, 208 and 209 scans for training, validation and testing, respectively. Using the same pre-processing operation as in [33]. The in-house dataset from Changhai Hospital includes 139 patients diagnosed with either SCN (Serous Cystic Neoplasm) or MCN (Mucinous Cystic Neoplasm). All volumes contain arterial phases (AP), portal venous phases (VP), and delayed phases (DP). Experienced radiologists annotated the pancreas and tumor regions in the volumes. The in-plane size of volumes is 512×512 with spacing ranges from 0.546 mm to 0.896 mm, and the number of slices ranges from 102 to 289. The volumes of three phases are co-registered into venous phase by Elastix [34] toolbox.

2) Implementation Details: The framework was implemented in Python with the PyTorch-based medical image framework MONAI and was trained and tested on a NVIDIA 4090 GPU. We set the learning rate to $1e-4$ and the number of epochs for model training to 200. Adopting the approach from [40], we first trained our model on single modality datasets using the trusted segmentation module to establish a robust baseline for each modality independently. After this initial stage, we engaged in advanced multi-modality training, combining the strengths of each individual modality to enhance the model's performance and accuracy. In training stage, we initially identify the volume's minimal bounding box and then proceed to randomly split it into a volume size of $128 \times 128 \times 128$. The data augmentation techniques are similar to [33]. In the test phase, we use the sliding window method with an overlap rate of 0.6. In our experiments, we use Dice score and 95% Hausdorff distance (HD95) to evaluate the segmentation results.

B. Comparisons With Other Methods

Our work presents a comparative analysis of the proposed REDNet model against well-established models. These include CNN architectures such as Attention U-Net [35], 3D U-Net [37], UNet++ [38], and transformer-based models like UNETR [36], as well as a hybrid model [33] that integrates both CNN and transformer techniques. Assessments also extend to various uncertainty-based models [7], [12], [39]. All the baseline models and our proposed model are trained with the same dataset split. The selection of the model was based on performance in the validation set. To be fair, all reported results are the direct output of the model without any post-processing.

1) Overall Evaluation: We carried out tests on both BRATS2021 and our in-house pancreas dataset. As shown in Table II, our segmentation strategy displayed outstanding performance without imperfection, outperforming most current technologies in fundamental metrics such as Dice and HD95. In the BraTS dataset, while it slightly trails the brain tumor-focused CKD method in some aspects, it outperforms other methods. On the in-house dataset, our method exceeds the performance of other models. Our in-house pancreas dataset includes real clinical data collected over 10+ years from different scanners, which better reflect actual clinical scenarios. Here, our method outperforms others because it is designed to handle imperfections.

2) Evaluation on Model Reliable With Imperfection: To assess the robustness of our proposed model against imperfection interference, we conducted experiments simulating a range of imperfect environments. Gaussian noise is a widely adopted synthetic imperfection for evaluating robustness in both general image processing and multi-modal frameworks. While real-world multi-modal data may exhibit diverse imperfections (e.g., motion artifacts, or modality-specific distortions). Gaussian noise serving as a standardized benchmark is widely used in research. Following [41], [42], and [43], we adopted the addition of Gaussian noise to the input data. The noise model is expressed mathematically as:

$$x_{noisy} = x + \mathcal{N}(0, \sigma^2), \tag{19}$$

where x represents the clean input data, while x_{noisy} denotes the corresponding data corrupted by additive noise. $N(0, \sigma^2)$ is the Gaussian noise generator with a mean of zero and a variance of σ^2.

As show in Fig. 2, the experiments are conducted on two datasets and each represented by two rows of images. Each dataset shows four different imperfection scenarios, each depicted in a respective graph for both Dice and HD95 metrics. Under noise-free conditions, our method might not outperform the CKD-TransBTS method, which employs clinical knowledge specifically designed for brain tumor segmentation. However, it still surpasses the accuracy of most compared methods. Importantly, our method demonstrates

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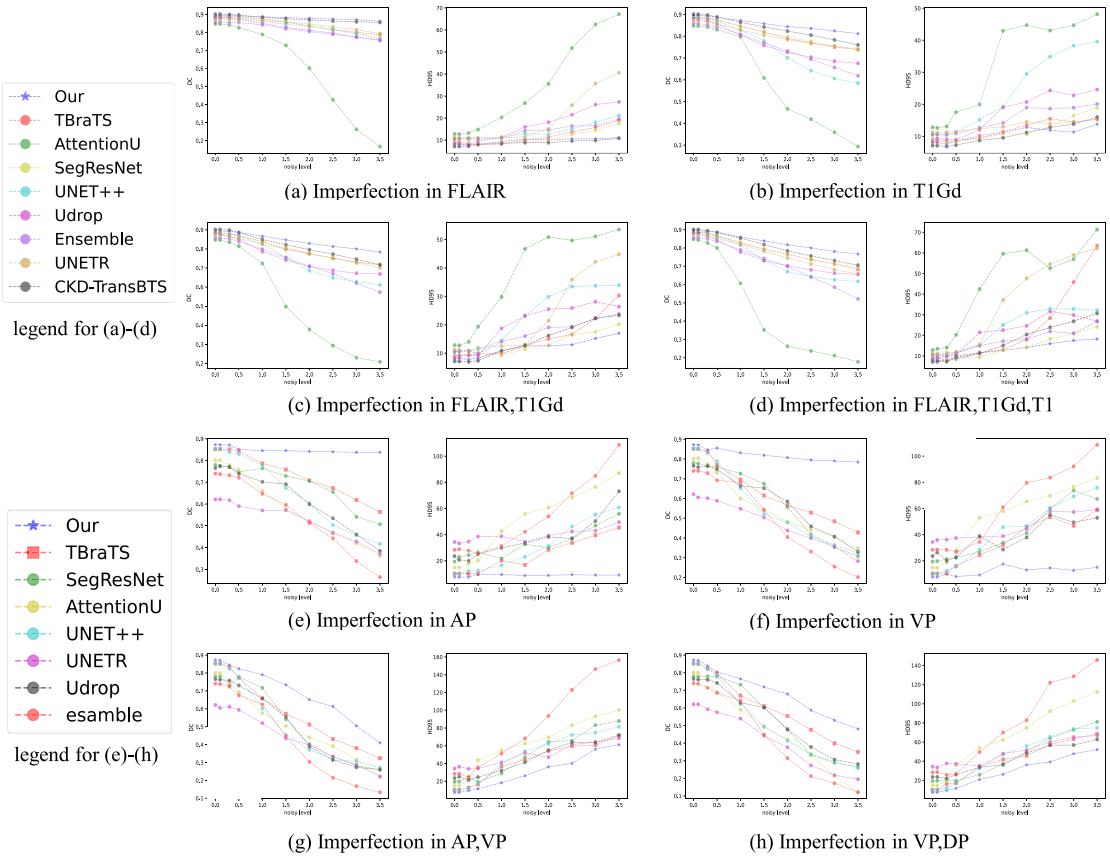


Fig. 2. The figure presents quantitative comparisons against varying imperfection magnitudes, with $\sigma = \{0,0.1,0.3,0.5,1,1.5,2,2.5,3,3.5\}$. Subfigures (a)–(d) display imperfection-affected images from the BRATS 2021 dataset across different modalities: (a) FLAIR; (b) T1Gd; (c) FLAIR and T1Gd; (d) FLAIR, T1Gd, and T1, respectively. Subfigures (e)–(h) depict the in-house dataset with noise introduced in sequences (e) AP; (f) VP; (g) AP and VP; (h) VP and DP.

superior performance in noisy conditions, consistently maintaining the highest Dice and the lowest HD95 scores, thereby outperforming all other methods in these scenarios. Our network is capable of efficiently extracting information from both single and multiple modalities, achieving superior segmentation accuracy that can be applied across various datasets.

We show the performances of our models against these perturbed scenarios. As show in Fig 3, we have visualized the segmentation performance of different models under imperfect conditions. We present cases of two patients, with each patient's data displayed in three rows. The first row illustrates the situation without noise, the second row shows noise added to the FLAIR, and the third row presents noise added to both the FLAIR and T1Gd. With the addition of noise, the accuracy of other methods' segmentation outcomes declines. While CKD-TransBTS achieves marginally higher metrics due to its domain-specific architectural priors, our framework prioritizes task-agnostic generalizability and reliability in imperfect scenarios. In the first case, with a larger tumor region, as the noise increases, UNET++ begins to produce excessive predictions, while SegResNet, TBraTS, Udrop, and UNETR mistakenly classify the regions. In the second case, with noise added, the UNet+ results progressively overestimate the area, while the SegResNet results shrink progressively until they are nearly non-existent. In contrast, our method maintains a robust performance and provides a consistently accurate representation.

Furthermore, while Gaussian noise is a widely adopted synthetic benchmark for methodological validation, we have also evaluated our method on two common imperfections in MRI: Rayleigh noise and intensity non-uniformity (INU). Rayleigh Noise: Noise in MRI images follows a Rayleigh distribution, particularly in low signal-to-noise ratio (SNR) regions [44]. To simulate this realistic condition, we introduced Rayleigh noise into the BraTS dataset. Intensity Non-Uniformity (INU): INU (also known as bias field) is a highly prevalent artifact in MRI [45]. It manifests as smooth, spatially varying intensity inhomogeneities, causing the same tissue type to exhibit different intensities across the image [46]. This artifact arises from factors like magnetic field or radio frequency (RF) inhomogeneity and patient anatomy [47]. We applied synthetic INU fields to the BraTS dataset to model this imperfection. We conducted experiments on the BraTS dataset under Rayleigh noise (scale=2) and INU conditions (magnitude=50). Our method was compared against eight well-performed models. Our method maintained high segmentation accuracy under both Rayleigh noise and INU corruption, outperforming all compared methods across the standard BraTS evaluation metrics.

Furthermore, we also selected imperfect data that actually exists in clinical practice to verify the effectiveness of our

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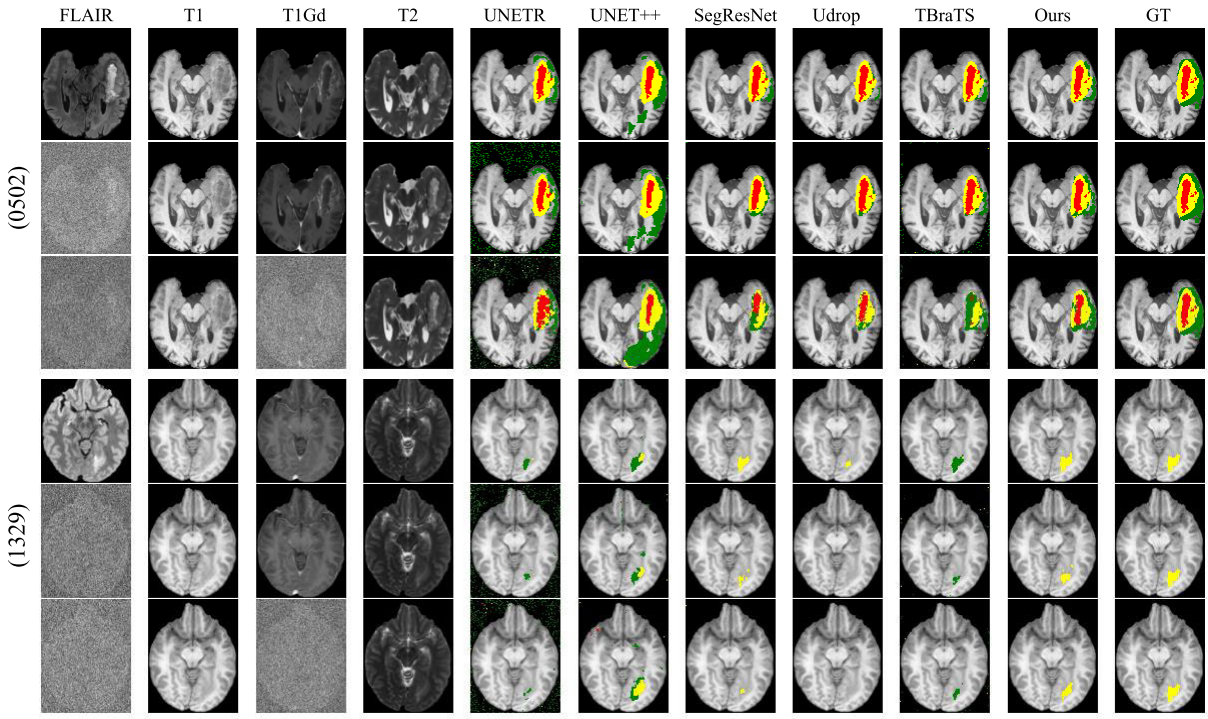


Fig. 3. Visualization of quantitative comparison of other methods on the BraTS21 dataset with noisy. Within each patient's three-row set, the first row depicts the noiseless scenario, the second showcases added noise in the FLAIR modality, and the third indicates noise addition in both FLAIR and T1Gd modalities. The level of noise $\sigma = 3.5$.

TABLE III
PERFORMANCE COMPARISON OF DIFFERENT MODELS ON THE BRATS21 DATASET UNDER RAYLEIGH NOISE AND INTENSITY NON-UNIFORMITY (INU) CONDITIONS

models	Rayleigh noise				Intensity non-uniformity			
	ET	TC	WT	Mean	ET	TC	WT	Mean
AttentionU	0.3883	0.4145	0.5961	0.4663	0.6856	0.7245	0.8499	0.7533
UNetr	0.6700	0.7852	0.8988	0.7847	0.6816	0.7420	0.8558	0.7598
SegResNet	0.6608	0.7098	0.7693	0.7133	0.6883	0.7441	0.8670	0.7666
UNet++	0.7409	0.8188	0.8471	0.8023	0.7197	0.7890	0.8823	0.7970
CKD-TransBTS	0.7299	0.7366	0.9068	0.7911	0.7516	0.8053	0.9203	0.8257
UE	0.7247	0.7765	0.8514	0.7842	0.6654	0.7257	0.8672	0.7528
DU	0.5548	0.7068	0.7665	0.6760	0.7042	0.7783	0.8844	0.7890
TBraTs	0.7037	0.7927	0.8760	0.7908	0.7199	0.7774	0.8891	0.7955
ours	**0.7850**	**0.8429**	**0.9083**	**0.8454**	**0.7646**	**0.8268**	**0.9249**	**0.8388**

model. In clinical image analysis, inherent and unforeseeable imperfections in data inevitably compromise segmentation accuracy. For instance, different CT scanners produce images with varying intensity distributions, leading to data heterogeneity that adversely affects segmentation precision. In our study, the majority of the imaging data was sourced from the Aquilion ONE device. As illustrated in Fig. 4, we select different anatomical position to show the results. The first patient images whose distribution closely matches the average distribution of our training data, resulting in optimal segmentation outcomes. Conversely, the second and third patients display images from the uCT 960+ and iCT 256 scanners, respectively, which deviate from the training data distribution, causing degradation in segmentation quality. When processing clinical pancreas data, our model leverages d_{in} and d_c to dynamically suppress the weights of bad modalities during multi-modal fusion, while enhancing the contribution of good modalities. This adaptive discounting mechanism ensures that the model automatically identifies and down-weights unreliable modalities caused by clinical imperfections (e.g., motion artifacts or scanner variations), while prioritizing high-quality imaging information. The inclusion of test data from uCT 960+ and iCT 256 scanners explicitly evaluates our method's robustness to real-world imperfections, such as cross-device intensity distribution shifts. While segmentation quality decreases for out-of-distribution scanners, the results remain anatomically consistent, underscoring the framework's clinical applicability.

This robustness to imperfection highlights the practical utility of our method in real-world clinical settings, where data quality cannot be guaranteed.

C. Ablation Studies

To better demonstrate the effectiveness across different modules, we chose the BRATS2021 dataset to simulate imperfections for our ablation experiments. Its higher quality allows

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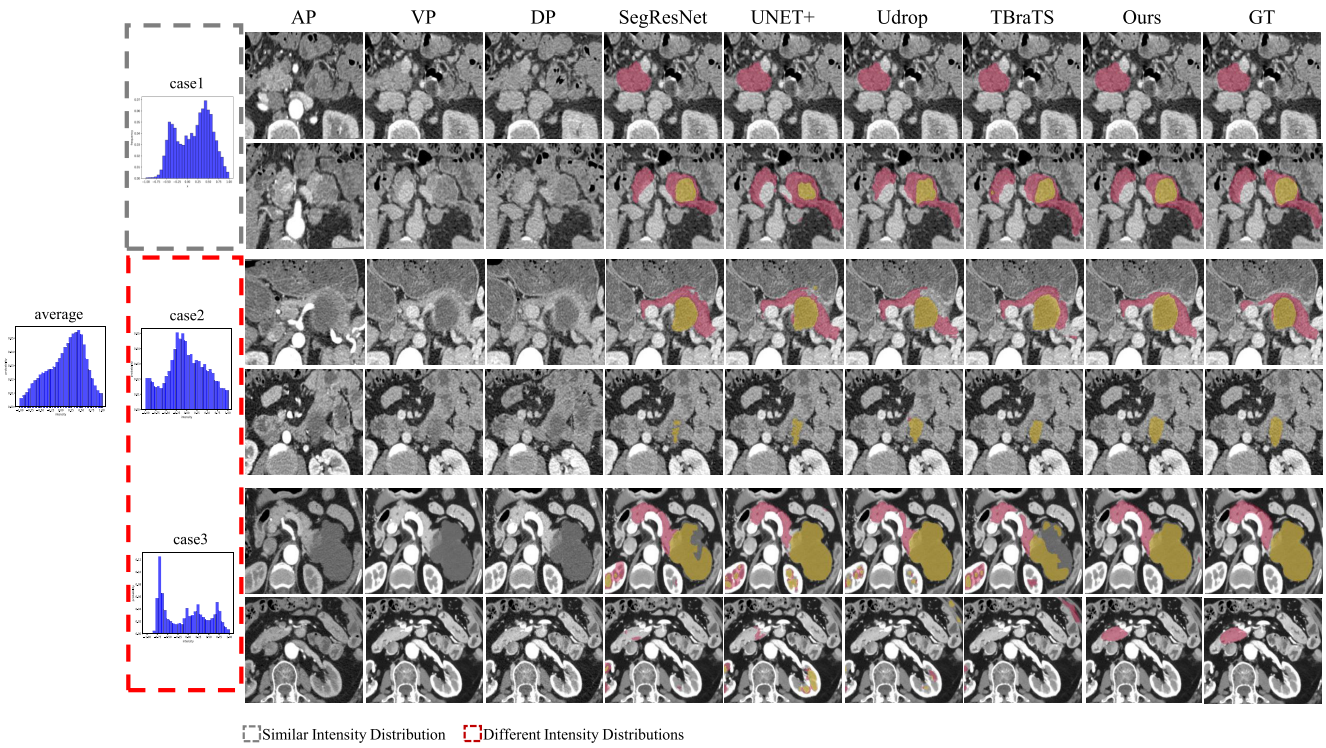


Fig. 4. This visualization compares segmentation results for inherent data imperfection. The first column shows the average intensity across all patients, and the second displays individual distributions for three patients. The intensity distributions represent whole-abdomen CT scans (not individual slices). Columns three to five present the actual patient images, while six to eleven illustrate segmentation results and labels from different methods. Each patient is represented by two rows: the first row corresponds to the pancreatic head region, and the second row to the pancreatic body region. The patient in the Rows 1-2, closest to the average distribution, shows optimal segmentation, while those farther from the average (Rows 3-6) yield poorer results.

TABLE IV

ABLATION STUDIES ON TRUST DISCOUNTING FUSION (F), INTRA-MODALITY CONSISTENCY EVALUATION MODULE (I) AND CROSS-MODALITY DIFFERENCE AGGREGATION MODULE (C). TEST WERE CONDUCTED BETWEEN EACH BASELINE MODEL AND OUR MODEL ON FOUR DIFFERENT IMPERFECTION CONDITIONS ("N-" IS USED TO DENOTE THE PRESENCE OF IMPERFECTION IN THE SPECIFIED MODALITIES)

Models	Ablation			Original			N-T1Gd			N-T1Gd-FLAIR			N-T1Gd-FLAIR-T1		
	F	I	C	ET	TC	WT	ET	TC	WT	ET	TC	WT	ET	TC	WT
1				0.7951	0.8654	0.9229	0.6760	0.7919	0.9015	0.6230	0.7330	0.8732	0.5787	0.6686	0.8365
2	✓			0.8455	0.8757	0.9295	0.6954	0.7905	0.9164	0.6531	0.7419	0.8775	0.6007	0.6778	0.8583
3	✓	✓		0.8142	0.8789	0.9314	0.6955	0.8107	**0.9197**	0.6587	0.7620	**0.8853**	0.6221	0.7221	**0.8750**
4	✓		✓	0.8521	0.8961	0.9337	0.6974	0.8016	0.9094	0.6767	0.7562	0.8713	0.6533	0.7324	0.8636
5	✓	✓	✓	**0.8532**	**0.8963**	**0.9343**	**0.7119**	**0.8136**	0.9109	**0.6892**	**0.7874**	0.8799	**0.6621**	**0.7665**	0.8691

for clearer evaluation of the impact of varying imperfection levels on different modules.

1) Effectiveness of Discounting Fusion Module: The Discounting Fusion Module is designed for the integration of outcomes from multiple modalities. Different from traditional fusion techniques, this module utilizes both intra-modal and cross-modal information to adjust results to improve segmentation.

Our ablation study aims to determine the significance of this module by evaluating it within both Model1 and Model2. Model1 employs DS evidence theory to directly fuse multimodal outputs from the evidential-based uncertainty estimation. In contrast, Model2 capitalizes on convolutional networks to first extract features from the data across different imaging phases before applying DS evidence theory for fusion. The F module then uses this information to selectively discount the data, which leads to enhanced segmentation accuracy. As displayed in Table IV, there is an improvement in the performance of Model2 compared to Model1. However, when confronted with imperfection conditions, the capacity to maintain this level of segmentation precision becomes more challenging.

2) Effectiveness of Intra-Modality Consistency Evaluation Module: The Intra-modality Consistency Evaluation Module (ICEM) plays a role in enhancing the model's ability to extract features from a single modality and offers better performance when imperfection affects a minority of the modalities. Our ablation study will assess its significance by evaluating model performance without this module.

The results show that the Intra-modality Consistency Evaluation Module (Model3) outperforms the baseline Model1 and Model2 across all imperfection scenarios. For example, under the N-T1Gd imperfection condition, Model3 shows enhanced performance with ET, TC, and WT scores of 0.6955,

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0.8107 and 0.9197 respectively, compared to the baseline scores of 0.6760, 0.8027 and 0.9099. This indicates the module's effectiveness in processing modality-specific information and mitigating the impact of imperfection. As imperfection affects more modalities, the ET scores decrease significantly, but the WT results remain relatively robust. For example, he ET score drops to 0.6221 under the N-T1Gd-FLAIR-T1 condition, while the WT score shows stability, decreasing slightly from 0.9314 to 0.8750. The module do better on extracting information from a single modality and minimizing the influence of imperfection data. This approach leads to impressive performance in handling imperfection within a single modality and in preserving the accuracy of whole tumor segmentation. However, as the imperfection increases, its reliability decreases when it comes to the detailed analysis of the core tumor structure.

3) Effectiveness of Cross-Modality Diversity Aggregator Module: To improve feature extraction and imperfection detection by cross-modality insights, we have developed a Cross-modality Difference Aggregation Module (CDAM). Our ablation study compared the performance of our model with and without this module.

Compared with Model1, Model2 and Model3, Model4 exhibits the highest segmentation accuracy among all models, with ET, TC, and WT scores of 0.8521, 0.8961, and 0.9337, respectively. This outcome shows the superior capability of the CDAM in extracting and integrating information across modalities, enhancing the accuracy of tumor segmentation. Under single modality imperfection conditions, specifically N-T1Gd, Model4 slightly underperforms Model3. However, as imperfection scenarios become more complex, as seen with N-T1Gd-FLAIR and N-T1Gd-FLAIR-T1, the benefits of the CDAM perform better. Model4 maintains robust performance in the face of complex imperfection, particularly in multimodal imperfection conditions, showing that the presence of the CDAM significantly contributes to the stability of the model's outputs.

D. Analysis of Model

We visualize intermediate results to analyze the effectiveness of our approach.

1) Imperfection-Aware: ICEM assesses the quality of testing data by quantifying its differences with the training data. The presence of imperfection tends to amplify this discrepancy. Imperfection is incrementally introduced into FLAIR images, and for each patient, the degree of diversity (θ) was calculated as described in Eq. (6). Subsequently, these values were plotted in a boxplot to visualize the variabilities. As show in Fig 5, the boxplot analysis reveals that our ICEM effectively detects discrepancies between test and training datasets with different imperfection levels. The central lines within each box, representing the median diversity values, exhibit a slight upward trend as imperfection levels increase, suggesting a rise in detected variances. Moreover, the expansion of the interquartile range (IQR) with increasing imperfection levels shows the increasing variability in data. As the data imperfection intensifies, this module performs well in capturing the level of discrepancy.

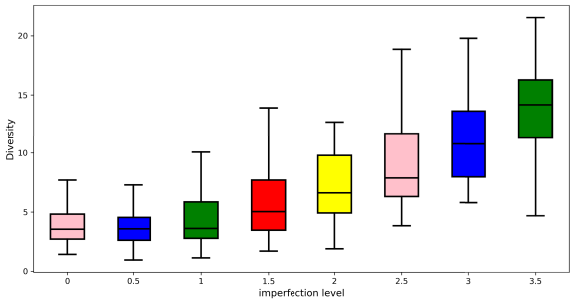


Fig. 5. Evaluating the link between diversity(θ in Eq. (6)) and imperfection on BraTS21. As imperfection increases, diversity grows accordingly.

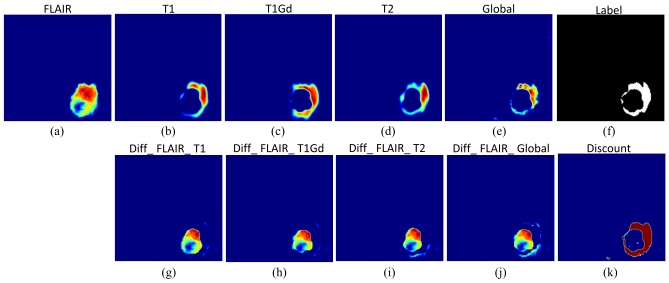


Fig. 6. The top row (a)–(e) visualizes the evidence from various modalities with (f) showing the corresponding ground truth labels. The second row (g)–(j) illustrates the discrepancy between the FLAIR modality and others, derived from Eq. (9), while (k) displays the outcome from Eq. (11), indicating the extent of discounting applied by the CDAM.

In CDAM, we acquire information disparity between the current modality and other modalities and harnessing it by neural network learning. This can discern areas of conflict when exposed to noise, thus offering discounts. Fig 6 presents the scenario where imperfection with $\sigma = 3.5$ is introduced into the FLAIR images. The first row shows the segmented results in different modalities by our network's branches and the ground truth. With the presence of imperfection, the evidence from the FLAIR modality worsens, while the data from other modalities remain unaffected, resulting in consistently good segmentation outcomes. The second row measures the discrepancy between the FLAIR modality and the others. These captured discrepancies are then processed through the neural network, which adeptly reduces the influence of areas where the modal information diverges substantially from that in other modalities. In the final discount map, sections likely to contain errors are intelligently discounted, as indicated by the color of red for areas with minimal adjustment and blue for those with greater discounting. This accurate modality-based discrimination enables the network to selectively apply corrections, mitigating the impact of inaccuracies in the segmentation results.

2) Uncertainty in Discounting: Our study utilizes the theory of discounting in subjective logic, applying a discount to mass and uncertainty values. The methodologies perform discounting based on the credibility of the information source. As depicted in the Fig. 7, the top row presents a depiction of the pre-discount state, while the second row illustrates the state post-discounting. In the initial undiscarded condition, we observe that the system predicts no results due to data

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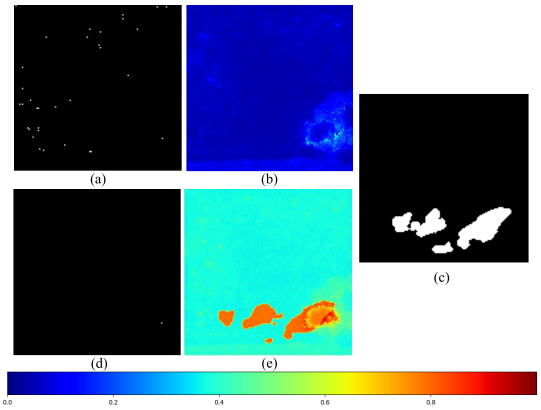


Fig. 7. The figure presents the impact of imperfection-induced discounting on FLAIR modality. Subfigures (a) and (b) show pre-discount predicted probability and uncertainty. Subfigures (d) and (e) illustrate post-discount probability and uncertainty, respectively. Subfigure (c) shows the segmentation labels.

corruption. However, because the damage is extensive, the uncertainty is quite low, which could potentially impair the fusion process. The second row reveals the situation after the discounting procedure has been applied. Even though the system's prediction is still absent due to the network issue, the corresponding prediction region exhibits a heightened level of uncertainty after the discounting adjustment. This adjustment leads to a better representation, reflecting the actual reliability of the corrupted data.

3) Discussion: The proposed method is designed to address general data imperfections by dynamically discounting unreliable modalities, rather than explicitly learning specific imperfection patterns. This method enables generalization to unseen imperfections, as it prioritizes modalities with higher agreement and suppresses those exhibiting deviations from the normal. Therefore, it demonstrates more robust performance when facing practical problems. While our discounting fusion module effectively suppresses unreliable modalities, the discounting operator scales down belief masses even for perfectly reliable data. Though the discount factor is close to 1, it still reduces the belief values. This always-on discounting slightly diminishes usable information from high-quality sources. We hope to explore more reasonable discounting mechanisms in future work.

V. Conclusion

In summary, the proposed REDNet exhibits robustness by addressing the issue of imperfection in multimodal medical image segmentation. Its reliable performance on both the BRATS2021 dataset and in-house datasets underscores its potential for clinical applications. With its combination of Intra-modality Consistency Evaluation Module and Cross-Modality Difference Aggregator Module and Discounting Fusion Module, REDNet dynamically adjusts the contributions of different modalities, enhancing the reliability in segmentation tasks.

Even when faced with challenging imperfection conditions, REDNet consistently outperforms traditional segmentation methods. Its ability to discern between reliable and imperfection data has significant implications for improving computer-aided diagnosis systems, paving the way for enhanced patient outcomes in clinical settings.

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